Employment Growth & Skill Demand Analysis (2024-2034) Using Python

1. Project Overview

This project analyzes **occupational employment trends and skill requirements** to understand how the workforce may change between **2024 and 2034**.The analysis combines **employment projection data** with **occupation and O*NET skill-related information** to identify occupations with strong future demand, occupations acing potential workforce decline, and the skills associated with different employment-growth patterns.The project follows a business-focused analytical approach, beginning with **data quality checks and Exploratory Data Analysis (EDA)** and progressing toward **workforce opportunity identification, skill-demand analysis, and workforce planning recommendations**.

2. Data Source

- **Dataset Name**: U.S. Bureau of Labor Statistics (BLS)
- **Source:** BLS Data Portal
- **File Type:** XLSX
- **Number of Records**: 145,000
- **Number of Attributes**: 13

The project uses two primary data sources: **U.S. Bureau of Labor Statistics (BLS)** for employment and 2024–2034 occupational projections, and **O*NET** for occupation-related skills and occupational characteristics. These sources were combined using occupational identifiers to support workforce and skill-demand analysis.

3. Problem Statement

Organizations need to understand how employment demand and skill requirements may change over time. This project addresses the challenge of identifying **future high-growth occupations, declining workforce areas, and in-demand skills** to support effective hiring, reskilling, and workforce development decisions.

4. Objectives:

- Identify **future high-growth occupations** based on projected employment growth.
- Identify **critical and declining workforce areas**.
- Analyze overall **employment growth patterns** from 2024 to 2034.
- Identify **skill demand across different growth patterns**.
- Analyze **skill requirements across occupations**.
- Generate **actionable workforce insights**.
- Support **workforce planning, hiring, reskilling, and development**.

5. Attribute (Column / Feature) Details:

Attribute	Description
Occupation	Name of the occupation.
Occupation_Code	Unique occupational classification code.
Employment_2024	Employment level in 2024.
Employment_2034	Projected employment level in 2034.
Employment_Change_24-34	Projected change in employment between 2024 and 2034.
Growth_Percent_24-34	Percentage growth in employment from 2024 to 2034.
EP skills ID	Identifier for the employment-projection skill record.
Skill	Skill associated with the occupation.
Skill_Score	Score representing the importance/level of the skill.

Attribute	Description
O*NET-SOC code	O*NET occupational classification code.
O*NET Element ID	Identifier for the O*NET occupational element.
O*NET element name	Name of the O*NET element.
O*NET data value	O*NET value associated with the occupational element.
Growth_Category *(Feature Engineered)*	Classifies occupations into **Declining, Stable, Growing, or High Growth** based on projected growth.
Skill_Level *(Feature Engineered)*	Categorizes skill scores into meaningful skill-level groups.
Employment_Size *(Feature Engineered)*	Categorizes occupations based on their employment size.
Future_Demand_Index *(Feature Engineered)*	Composite indicator created to represent the future workforce demand of an occupation using employment and growth-related factors.

6.Tools & Technologies

Python – Data cleaning, feature engineering, EDA, statistical analysis, and visualization
Pandas & NumPy – Data manipulation and numerical analysis
Matplotlib & Seaborn – Data visualization
Power BI – Interactive dashboards and business reporting
SQL / MySQL – Data querying, transformation, and analysis
Jupyter Notebook – Data analysis and experimentation
Git & GitHub – Version control and project management
Generative AI / LLMs – Analytical assistance and insight generation
AI Agents / Agentic AI – Future enhancement for automated workforce analysis and decision-support workflows
n8n – Future workflow automation and integration
APIs – Future integration with external workforce and skills data sources
Automation & Scheduled Workflows – Future automated data refresh, analysis, and reporting
Power BI Service – Future dashboard publishing, sharing, and scheduled refresh

7.Data Pre-Processing and Feature Engineering

- Checked dataset **structure, data types, missing values, duplicates, and unique values**.
- Validated numerical fields such as **employment and growth metrics**.
- Reviewed occupation and skill records to ensure the data was suitable for analysis.
- Calculated **Growth Category** to classify occupations as Declining, Stable, Growing, or High Growth.
- Created **Skill Level** based on skill scores.
- Created **Employment Size** to classify occupations based on workforce size.
- Created **Future Demand Index** to represent future workforce demand using employment and growth factors.
- Prepared the engineered features for **EDA, workforce opportunity analysis, skill-demand analysis, and business decision-making**.

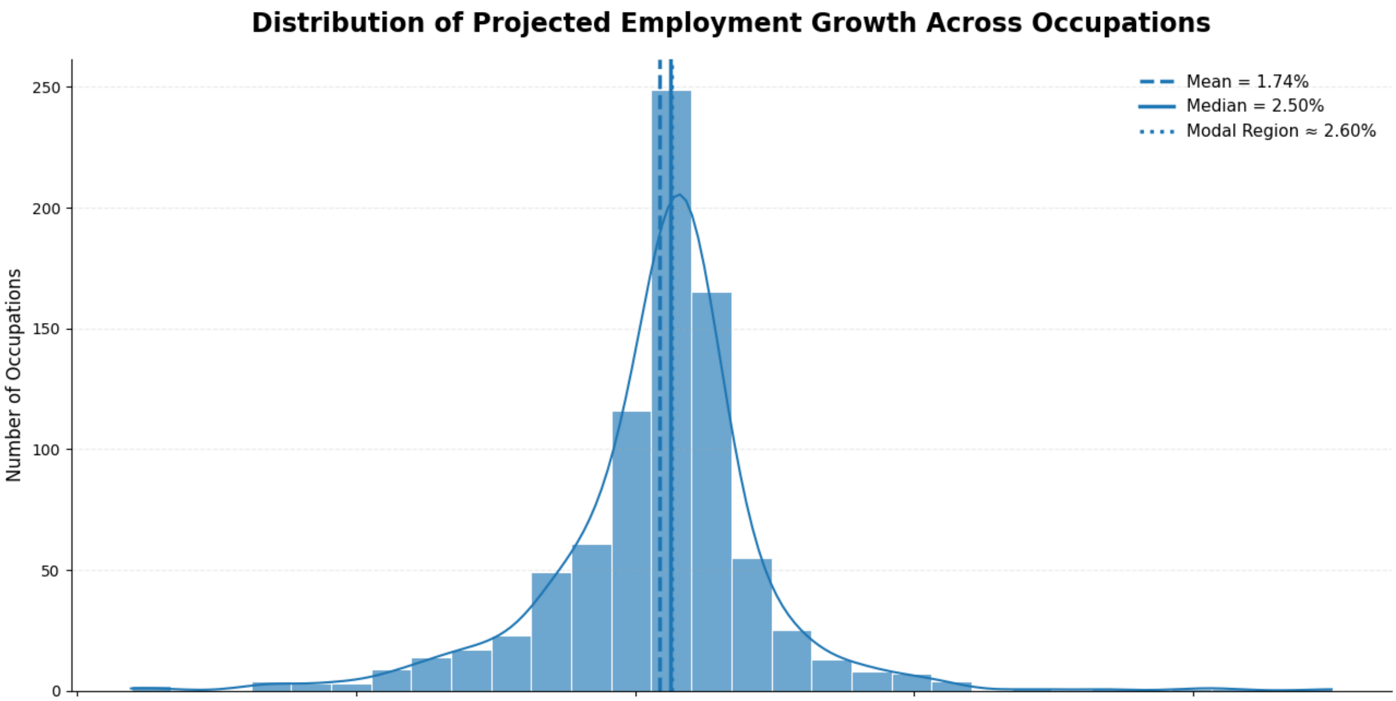
Feature Engineering

- **Growth_Category** – Categorized occupations into Declining, Stable, Growing, and High Growth based on projected employment growth.
- **Skill_Level** – Classified skills into meaningful levels based on `Skill_Score`.
- **Employment_Size** – Classified occupations based on their 2024 employment level.
- **Future_Demand_Index** – Created a composite indicator using employment size and projected growth to identify occupations with stronger future workforce demand.

8. Statistical Analysis – Measure of Central Tendency

- The projected employment growth was analyzed using **mean = 1.74%, median= 2.50%, minimum, and maximum** values to understand the central tendency and overall range of employment growth across occupations.
- A **histogram** was used to visualize the distribution and identify how occupations are concentrated across different growth levels.

Figure 1:



Interpretation

The histogram shows the distribution of projected employment growth across occupations, with the mean, median, and modal region highlighted. Comparing these measures helps identify the central tendency and whether employment growth is symmetrically distributed or influenced by occupations with unusually high or low projected growth.

Key insight

The relationship between the mean, median, and mode indicates the underlying shape of the employment-growth distribution. A noticeable separation between these measures suggests skewness, while close alignment indicates a more concentrated and symmetric growth pattern.

9. Data Visualization:

1. High-Growth Occupations: Current vs Future Employment

Type: Multivariate dumbbell-style visualization

Current → Future Workforce Expansion

Interpretation: The chart shows how employment is expected to change from 2024 to 2034 among high-growth occupations. The distance between the current and future employment points represents the expected workforce expansion.

Key insight: The occupations with the largest absolute employment increase will potentially require the greatest amount of future workforce capacity.

Observation: Look for the occupations with the longest expansion distance between 2024 and 2034. These are more important for workforce planning than occupations with high percentage growth but relatively small employment bases.

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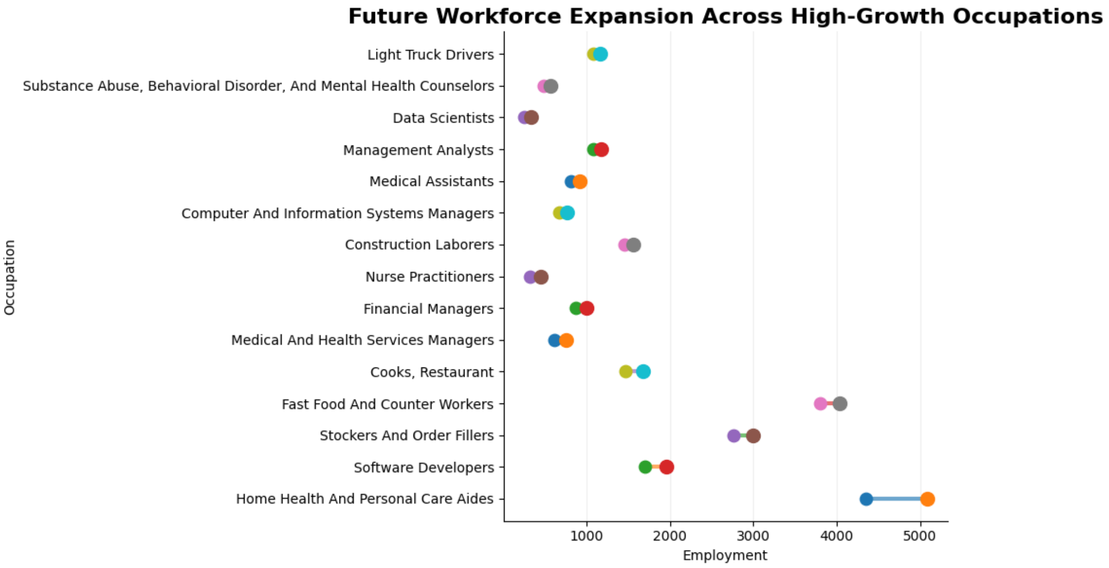

2. Critical Workforce

Workforce Criticality Matrix

Type: Multivariate scatter plot

Which occupations deserve priority because their workforce size × projected change creates substantial workforce exposure?

Interpretation: The chart evaluates occupations using two dimensions: current workforce size and projected employment change.

Key insight: Large occupations experiencing substantial workforce change have greater strategic workforce significance because changes in these occupations can affect a larger number of workers.

Observation: Pay particular attention to occupations in the high-employment region with significant positive or negative employment change. These represent occupations where workforce planning can have the largest impact.

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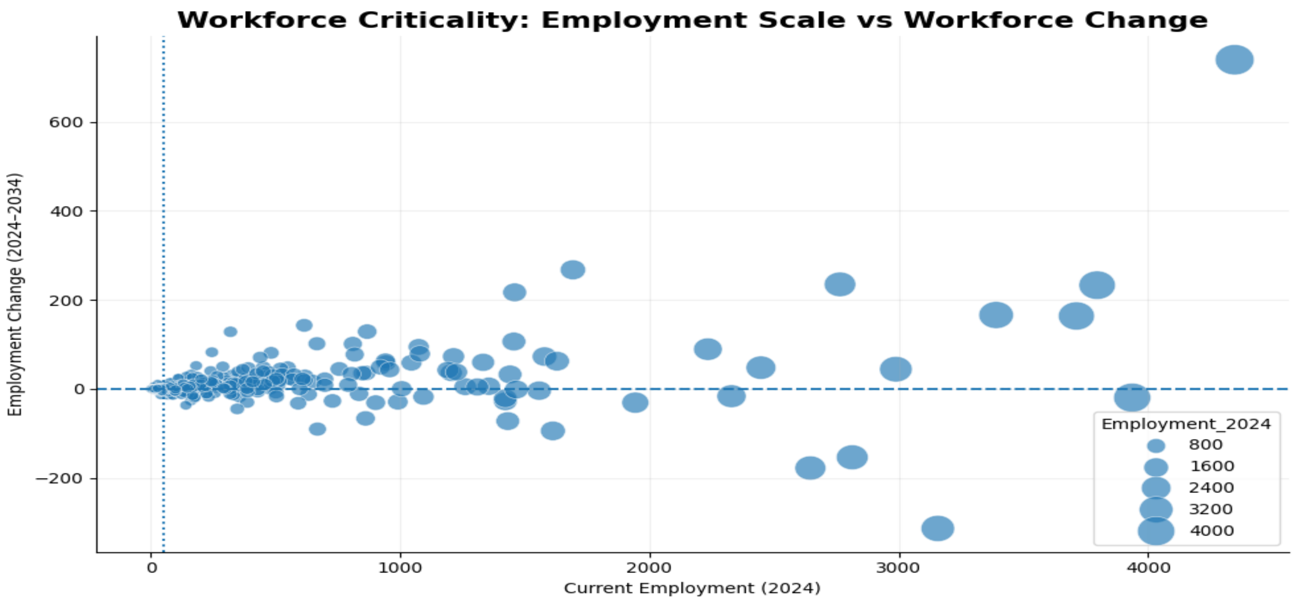


Critical Workforce Exposure

Type: Bivariate ranking

Workforce Exposure

Interpretation: The Workforce Exposure Index combines the size of an occupation's current workforce with the magnitude of its projected growth or decline.

Key insight: An occupation with a large workforce can have high workforce exposure even when its percentage growth is relatively moderate.

Observation: The highest-ranked occupations represent areas where changes in employment could affect a substantial workforce population and therefore deserve closer planning attention.

3. Employment Growth Patterns

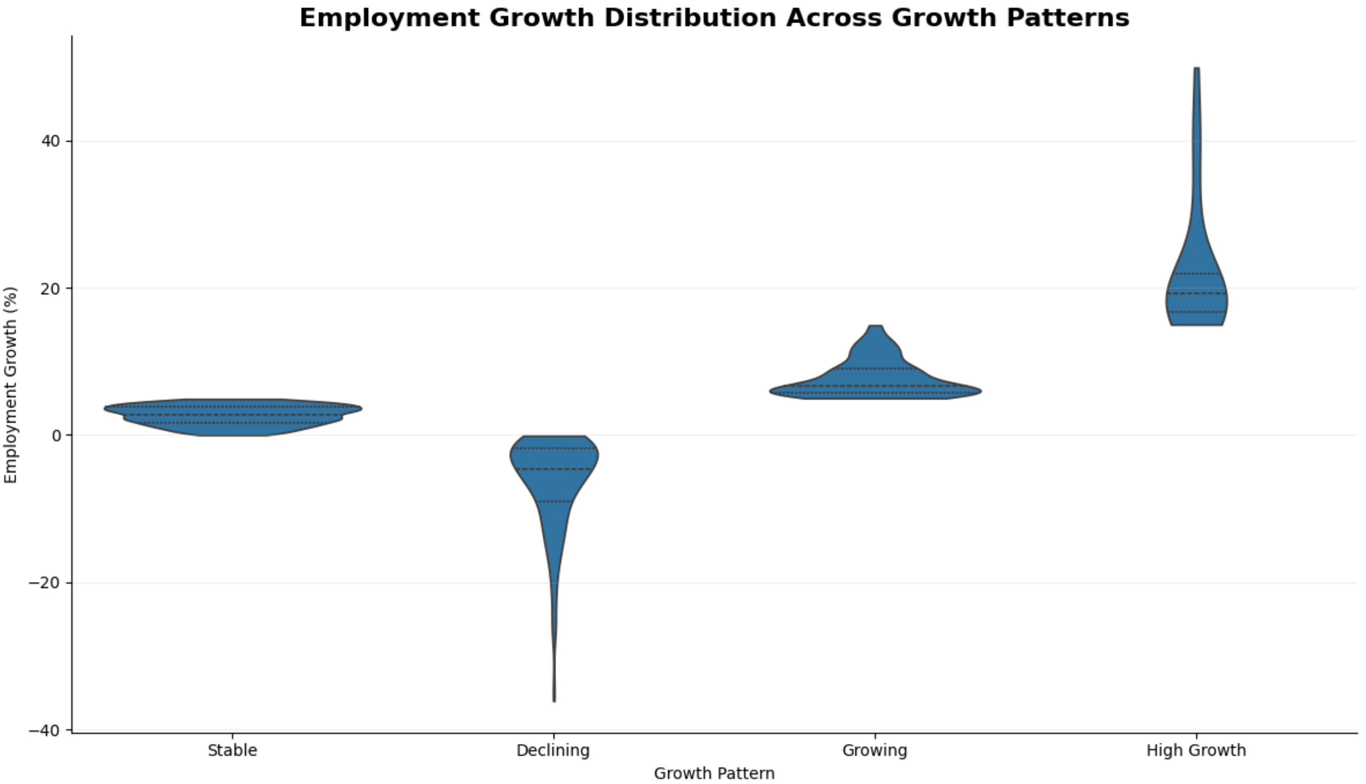
Type: Multivariate box/violin visualization

Growth Distribution Across Patterns

Interpretation: The violin plot compares the distribution of employment growth across the four growth categories: Declining, Stable, Growing, and High Growth.

Key insight: The distributions show how employment growth varies across workforce patterns rather than treating all occupations as having the same growth behavior.

Observation: Look at the median, spread, and overlap of each category. A wider distribution indicates greater variation among occupations within that growth pattern, while isolated values indicate potential outliers.

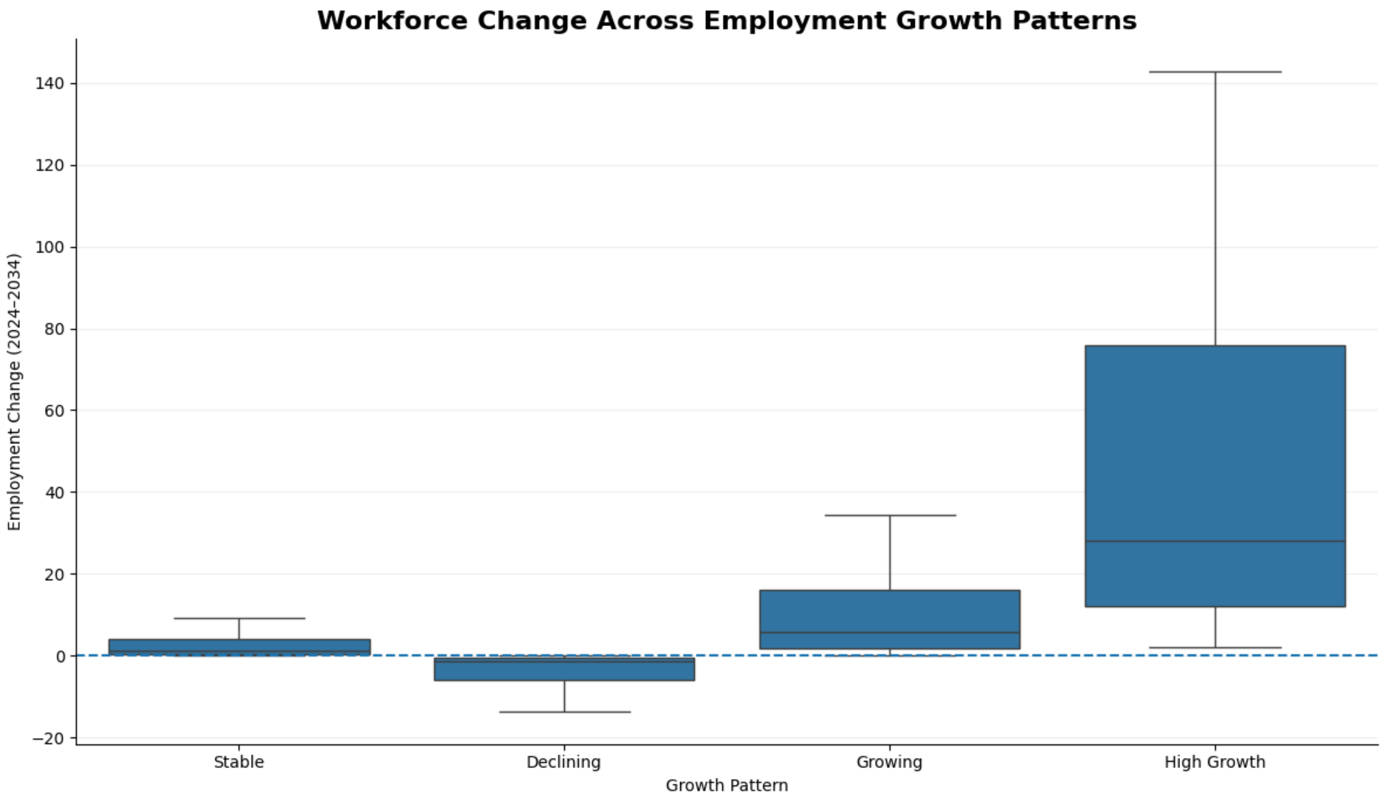


Employment Change Across Growth Patterns

Interpretation: This chart compares the absolute employment change associated with each growth pattern.

Key insight: Percentage growth and absolute workforce change are different dimensions. A growth category can contain occupations with very different numbers of additional or lost jobs.

Observation: The High Growth group should generally show positive employment change, while the Declining group should show negative change. The magnitude and spread indicate how strongly each pattern affects workforce size.



4.Skill Demand Across Patterns

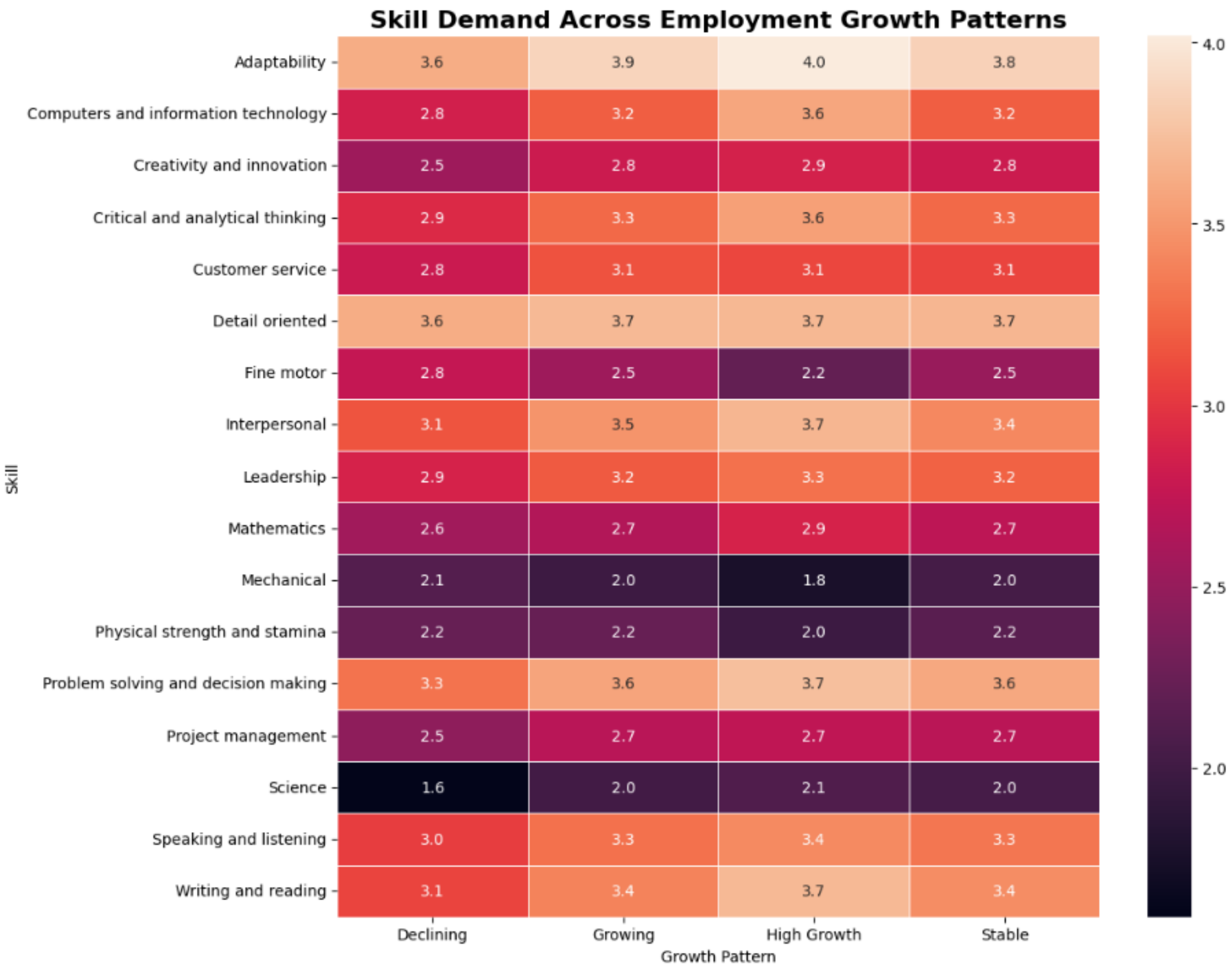
Skill Demand Heatmap

Type: Multivariate heatmap

Interpretation: The heatmap compares average skill scores across employment-growth patterns. Darker/higher-value cells indicate stronger skill demand within a particular growth pattern.

Key insight: The visualization identifies skills whose importance is concentrated in growing and high-growth occupations, helping distinguish future-oriented skills from skills associated mainly with stable or declining occupations.

Observation: Look for skills with consistently high scores in the High Growth column. Skills that remain strong across several growth patterns are broadly transferable, while skills concentrated in one pattern are more specialized.

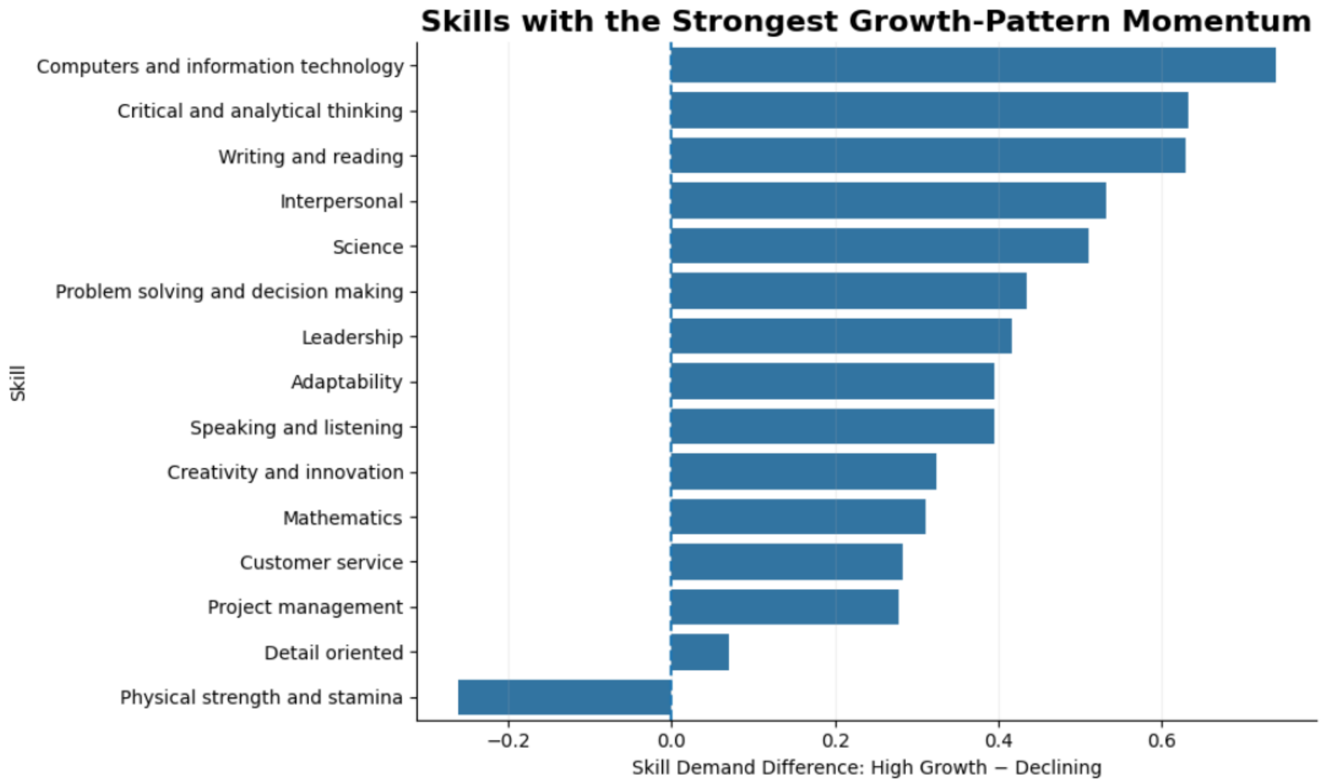


Skill Demand Momentum
Type: Bivariate ranking

Interpretation: Skill Momentum measures the difference between a skill's average demand in High Growth occupations and its demand in Declining occupations.

Key insight: A positive momentum score indicates that the skill is more strongly associated with high-growth occupations and can therefore be considered a potential future-oriented skill.

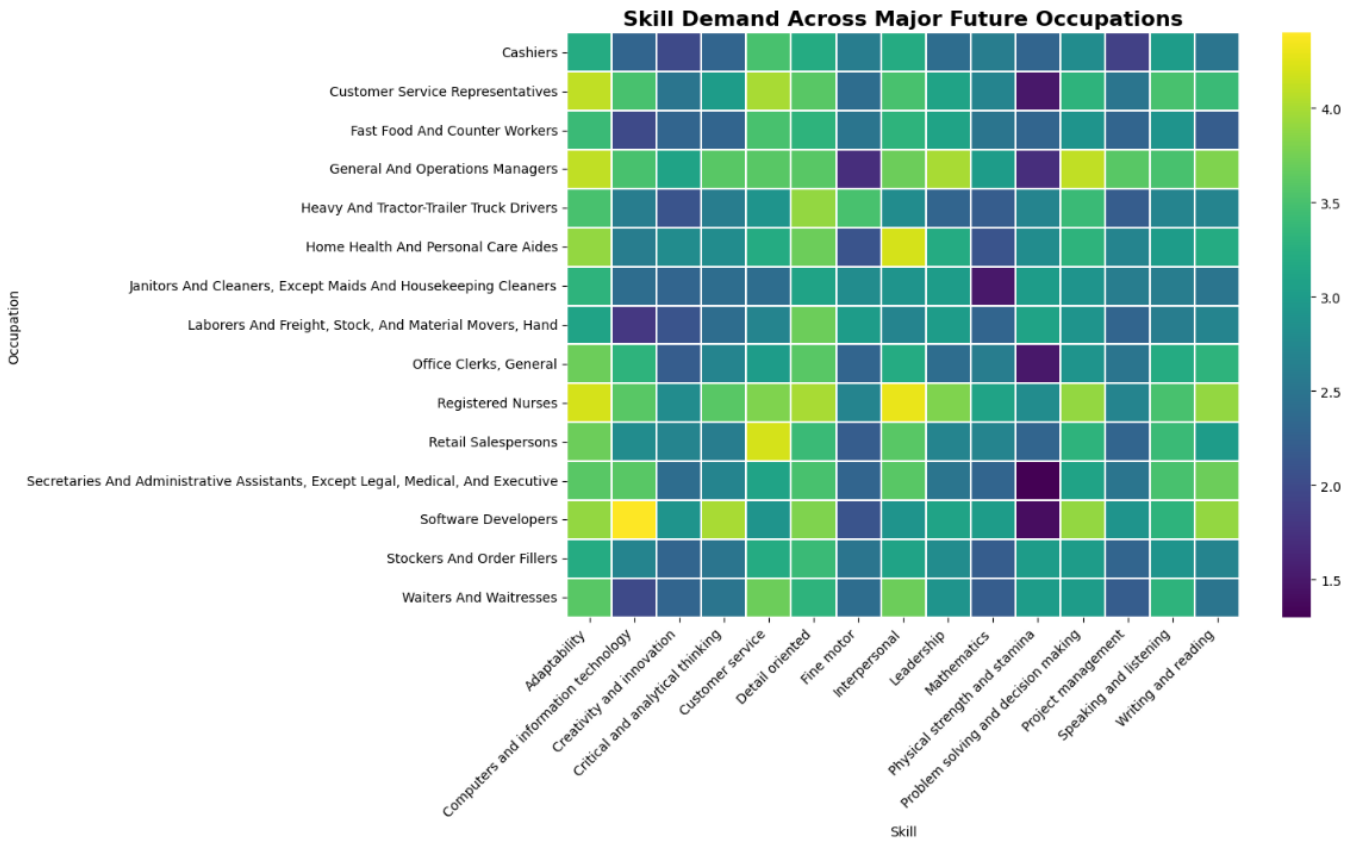
Observation: The skills at the top of the chart have the largest positive difference between High Growth and Declining occupations. These are strong candidates for upskilling and workforce-development programs.



5. Skill Demand Across Occupations

Occupation–Skill Demand Heatmap

Type: Multivariate heatmap



Business insight & business-side interpretation

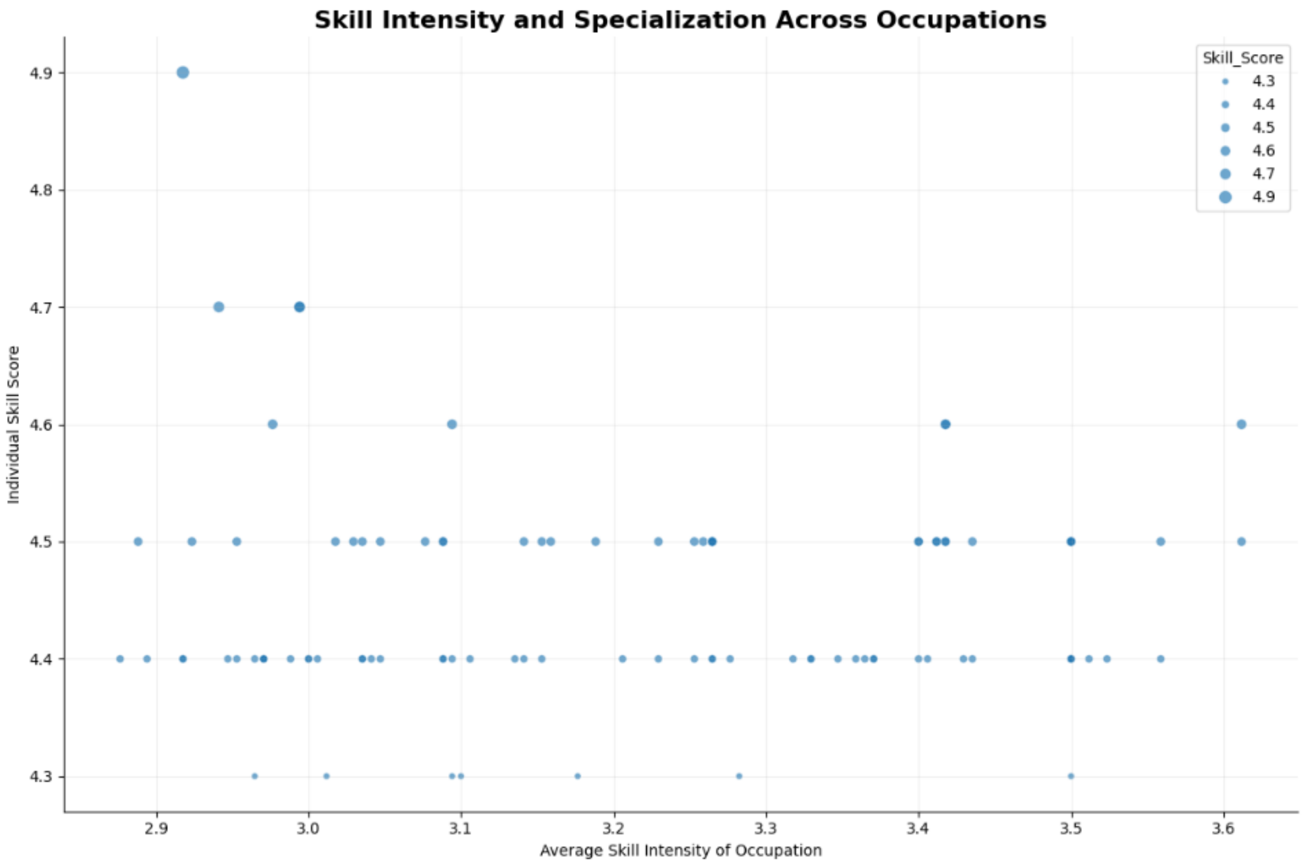
This **heatmap** compares skill demand scores across **15 major future occupations and 15 skills**, where darker shades represent lower scores and brighter shades represent higher scores.

- **Registered Nurses** show consistently high skill scores, with several skills around **3.5–4.3**, indicating broad and strong skill requirements.
- **Software Developers** show particularly high demand for **Computers and Information Technology (~4.3)** and **Critical and Analytical Thinking (~4.0)**.
- **General and Operations Managers** show strong requirements in **Adaptability (~4.2), Leadership (~4.0), and Project Management (~4.0)**.
- **Customer Service Representatives** show high scores for **Adaptability (~4.2)** and **Customer Service (~4.2)**.
- Lower values around **1.4–2.0** indicate comparatively lower skill requirements for those occupation–skill combinations.

Key insight: Some skills appear across many occupations, while others have high importance only within specific occupations. This distinguishes transferable skills from occupation-specific skills.

Observation: Look for columns with consistently high values across many occupations. These represent skills with broader workforce applicability. Isolated high-value cells indicate specialized skill requirements.

Skill Specialization Across Occupations
Type: Multivariate scatter



Business insight & business-side interpretation

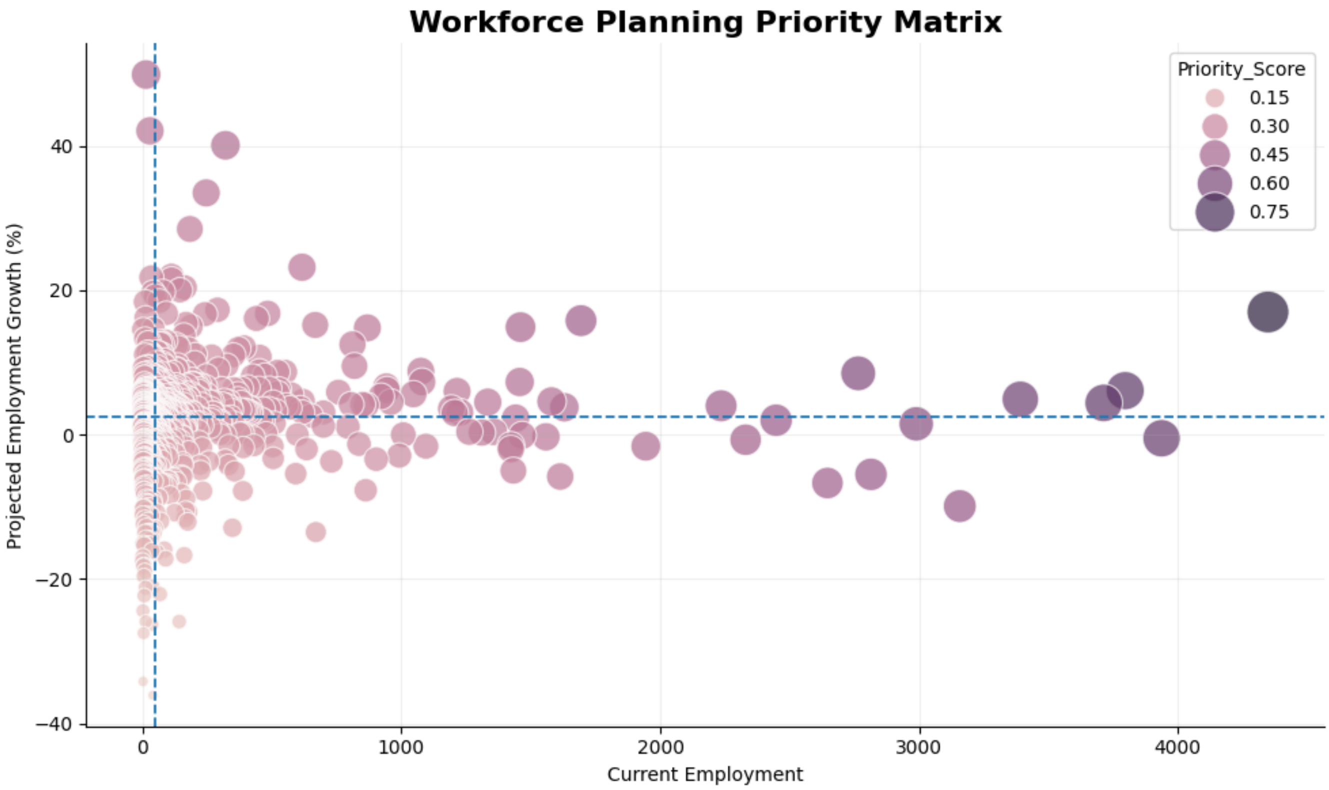
This **scatter plot** shows the relationship between **Average Skill Intensity of an occupation** and its **Individual Skill Score**. The average skill intensity ranges approximately from **2.88 to 3.62**, while individual skill scores range from **4.3 to 4.9**.

- Occupations around **3.4–3.6 average skill intensity** still show high individual skill scores of around **4.5–4.6**, indicating strong skill requirements even at higher overall intensity.
- The **highest individual skill score is 4.9**, occurring at an average skill intensity of approximately **2.92**.
- Several occupations around **3.0–3.2** have skill scores between **4.4 and 4.7**, showing that high individual-skill requirements can exist even when overall skill intensity is moderate.
- Most observations are concentrated around **4.4–4.5 skill scores**, suggesting that these are the most common skill-intensity levels in the analyzed occupations.
- The points increase in size with higher **Skill_Score**, so larger bubbles highlight occupations/skills with stronger individual skill requirements.

Key business insight

There is **no simple one-to-one relationship** between overall occupation skill intensity and individual skill score. Some occupations with moderate average intensity still require very high individual skills. Therefore, workforce development should consider **both overall occupation intensity and specific skill requirements** when prioritizing training and hiring.

6. Actionable Insights for Workforce Planning & Development
Workforce Priority Matrix
Type: Multivariate scatter



Business insight & business-side interpretation

This **bubble scatter plot** represents the **Workforce Planning Priority Matrix**, comparing **Current Employment** with **Projected Employment Growth (%)**. Bubble size represents the **Priority Score**.

- **Current employment:** ranges from approximately **0 to 4,400**.
- **Projected growth:** ranges from approximately **−37% to +50%**.
- The horizontal reference line is around **2.5% growth**, separating lower-growth from higher-growth occupations.
- The vertical reference line is around **0 current employment**, establishing the workforce-size reference point.
- The strongest visible priority is the large bubble near **4,300 current employees and ~17% projected growth**, indicating a **large workforce combined with strong growth**.
- Several occupations around **2,700–3,800 current employment** show approximately **5–10% growth**, representing substantial workforce-planning opportunities.

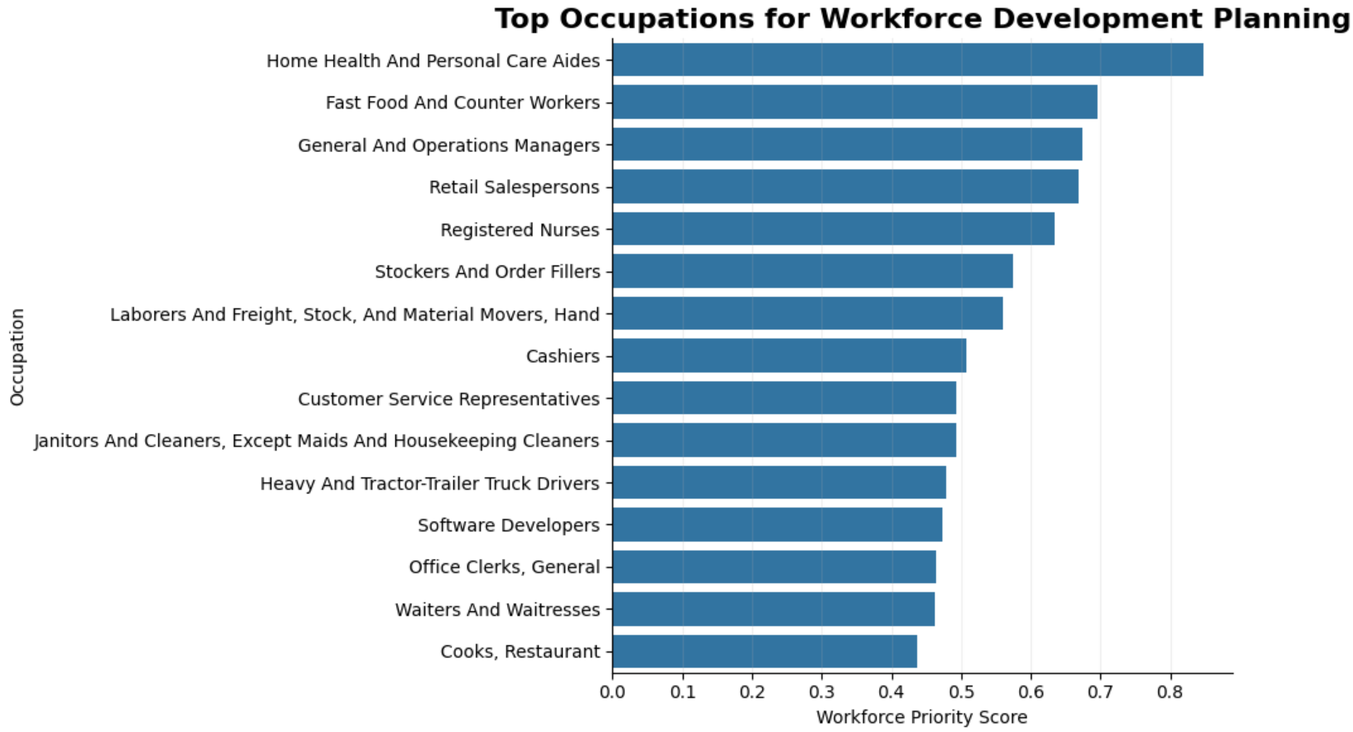
- Occupations below **0% growth**, including values near **−20% to −35%**, indicate potential **workforce decline and reskilling/transition needs**.
- The largest bubbles have Priority Scores approaching **0.75**, indicating the highest priority according to your scoring model.

Key business insight:

The most important workforce-planning targets are occupations combining **large current employment + positive projected growth + high Priority Score**. These areas may require greater investment in **hiring, talent development, and capacity planning**, while negative-growth occupations require **reskilling or workforce-transition strategies**.

Workforce Development Priority

Type: Multivariate ranking



Business insight & business-side interpretation

This **horizontal bar chart** ranks the **top 15 occupations for workforce development planning** based on the **Workforce Priority Score**.

- **Home Health and Personal Care Aides** has the highest priority score at approximately **0.85**, making it the strongest workforce-development priority.
- **Fast Food and Counter Workers** follows at approximately **0.70**.
- **General and Operations Managers** and **Retail Salespersons** are both around **0.67–0.68**.
- **Registered Nurses** score approximately **0.63**, indicating a relatively high workforce-planning priority.
- The remaining occupations have scores roughly between **0.44 and 0.58**.

Key business insight: The analysis identifies occupations where workforce development resources can be prioritized. The highest-scoring occupations should receive greater attention for **hiring, training, skill development, and workforce capacity planning**.

11. Advanced Analytics Insights

Descriptive Analysis

1. Data Overview

Examined the structure, size, data types, missing values, duplicates, and key attributes of the workforce dataset.

2. Employment Analysis

Analyzed 2024 employment, projected 2034 employment, and absolute employment change to understand the overall workforce size and expected changes.

3. Growth Analysis

Analyzed projected employment growth percentages to understand the distribution of declining, stable, growing, and high-growth occupations.

4. Skill Analysis

Examined skill scores and O*NET elements to understand the skill requirements associated with different occupations.

Diagnostic Analysis

1. Growth Pattern Analysis

Compared employment growth across occupations to identify areas of workforce expansion and decline.

2. Growth Category Analysis

Classified occupations into Declining, Stable, Growing, and High Growth categories to understand the underlying workforce pattern.

3. Skill–Occupation Analysis

Examined how skill requirements differ across occupations and identified occupations with higher skill intensity.

4. Workforce Opportunity Analysis

Compared current employment with projected growth to identify high-opportunity, stable, emerging, and potential-risk workforce areas.

Predictive Analysis

1. Prediction Objective

Predicted the likely growth category of occupations using employment, growth, skill, and engineered workforce features.

2. Feature Engineering

Created Growth Category, Skill Level, Employment Size, and Future Demand Index to provide meaningful inputs for predictive modeling.

3. Predictive Modeling

Applied classification models to identify whether occupations are likely to be Declining, Stable, Growing, or High Growth.

4. Business Application

The predictions can support future hiring priorities, workforce capacity planning, reskilling, and skill-development strategies.

Prescriptive Analysis

1. High-Growth Workforce Planning

Prioritize high-growth occupations for recruitment, workforce expansion, and talent development.

2. Critical Workforce Management

Identify declining occupations and develop appropriate reskilling, redeployment, or transition strategies.

3. Skill Development

Prioritize skills associated with growing occupations for targeted upskilling and training programs.

4. Workforce Investment

Use the Workforce Priority Score and Future Demand Index to allocate workforce-development resources toward occupations with greater strategic importance.

12. Project Extension – Automation Workflow

1. Overview

This project analyzes projected U.S. employment trends from **2024 to 2034** at the occupation level. The original dataset contained detailed occupation, skills, and O*NET element records. The data was transformed into an occupation-level dataset containing employment figures, employment change, growth percentage, growth category, employment size, and a future employment index.

The project uses **n8n automation** to process the data and an **AI Agent** to answer business questions using the resulting Master Dataset stored in Google Sheets.

2. Automation Workflow Explanation

The n8n workflow automates the transformation of occupation-level employment data into an analysis-ready Master Dataset.

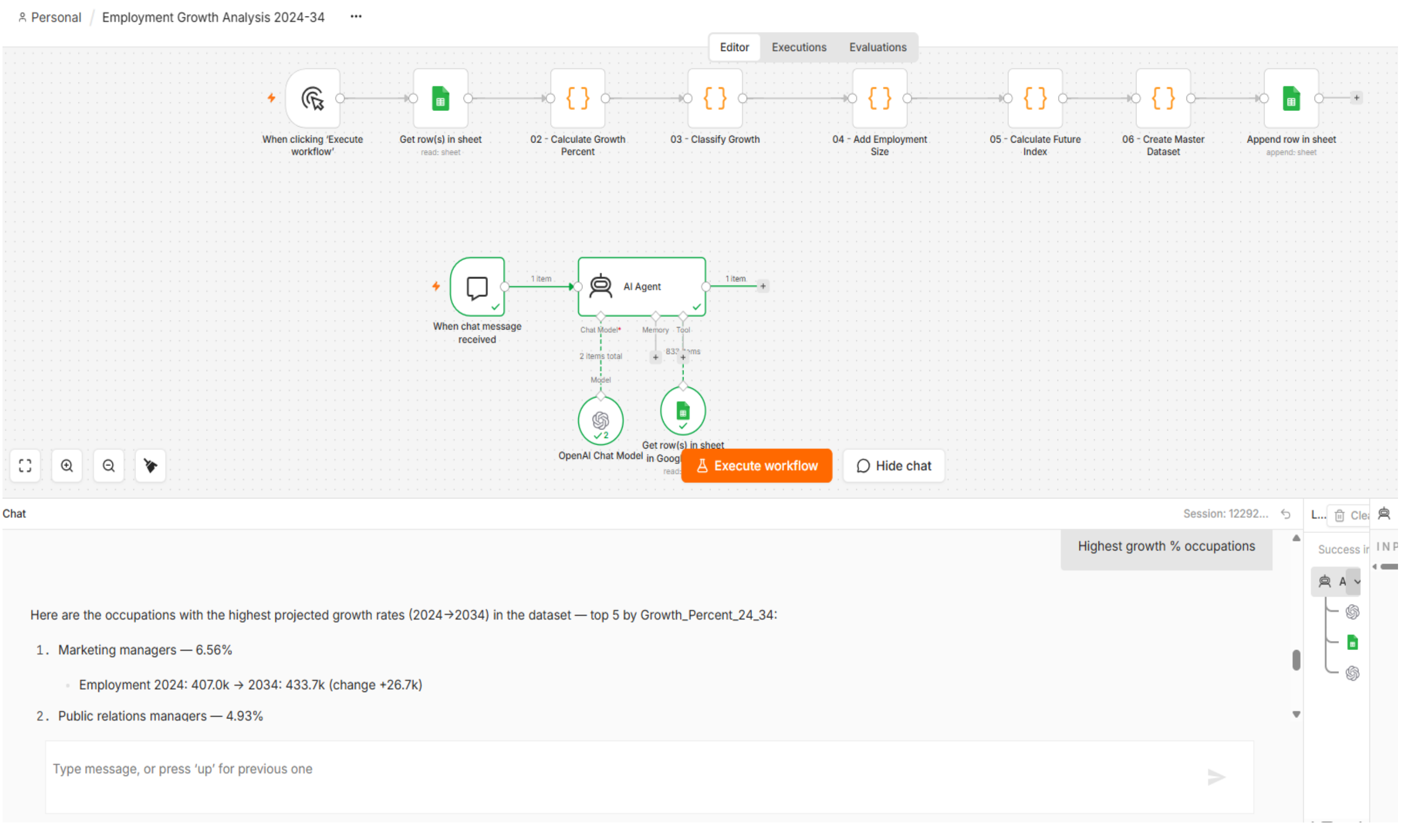
Workflow:

Google Sheets → Calculate Growth % → Classify Growth → Add Employment Size → Calculate Future Index → Create Master Dataset → Google Sheets

Step-by-step

- **Get Row(s) in Sheet** – Reads the occupation-level employment data from Google Sheets.
- **Calculate Growth Percent** – Calculates employment growth percentage from 2024 to 2034.
- **Classify Growth** – Categorizes occupations as Declining, Stable, Growing, or High Growth.
- **Add Employment Size** – Groups occupations based on their 2024 employment size.
- **Calculate Future Index** – Calculates the ratio of projected 2034 employment to 2024 employment.
- **Create Master Dataset** – Selects and organizes the required analytical fields.
- **Google Sheets** – Stores the resulting Master Dataset for analysis and AI Agent access.

The AI Agent is connected separately to the Master_Dataset Google Sheet so users can ask natural-language questions about employment trends.



3. Key Features

- **Automated data transformation** using n8n
- **Occupation-level employment analysis**
- **Growth percentage calculation**
- **Growth categorization**
    - Declining
    - Stable
    - Growing
    - High Growth
- **Employment size segmentation**
    - Small
    - Medium
    - Large
    - Very Large
- **Future Employment Index**
- **Centralized Master Dataset** in Google Sheets
- **AI-powered natural-language analysis**

- Ability to answer business questions such as:
  - Which occupations have the highest employment growth?
  - Which occupations are declining?
  - Which occupations have the largest number of jobs added?
  - How does one occupation compare with another?

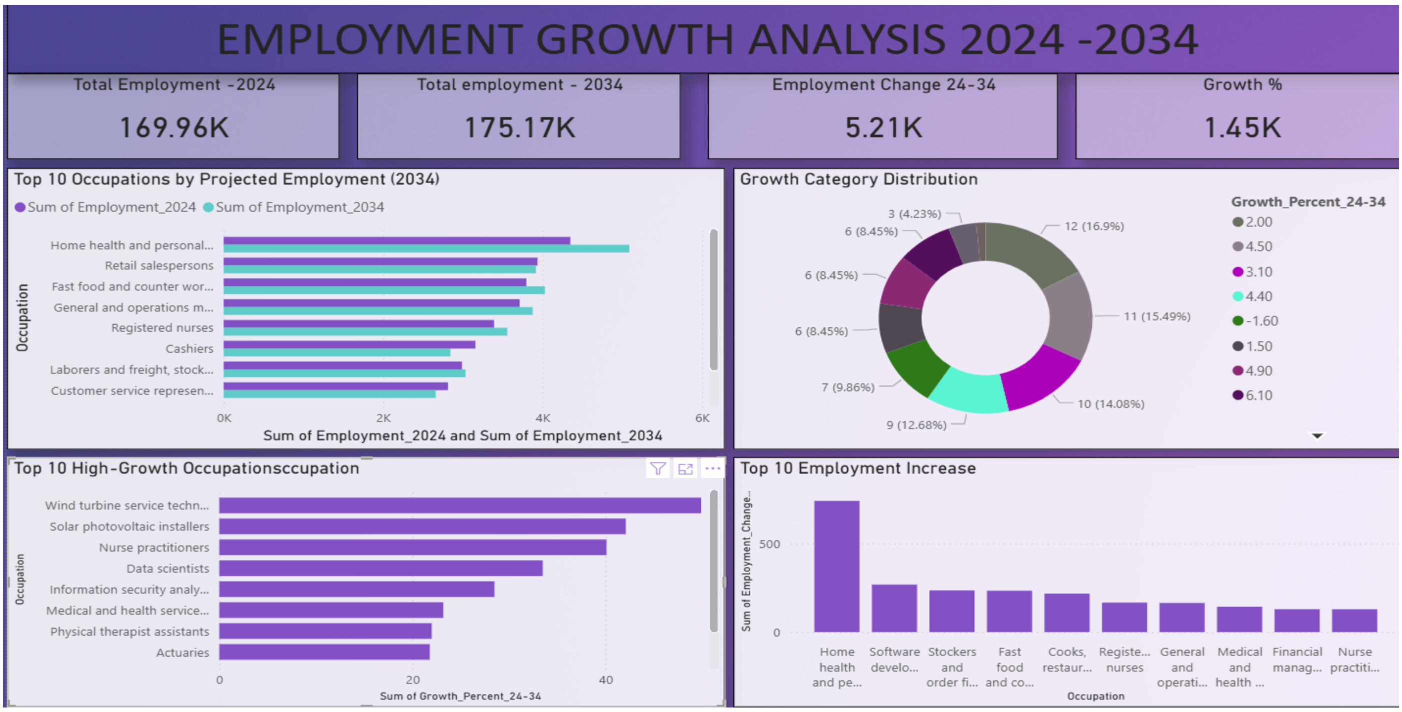
This is a good **first documentation section**. Next, we should document the **Feature Engineering / Calculations** because that explains exactly how each of those analytical fields was created.

Tools Used

Tool	Purpose
Google Sheets	Store the occupation-level source data and the final Master_Dataset
n8n	Automate the complete data-processing workflow
JavaScript / Code Nodes	Perform calculations and feature engineering
OpenAI GPT-5 Mini	Power the AI Agent for natural-language analysis
n8n AI Agent	Answer business questions using the Master_Dataset
Power BI	Create interactive dashboards and visualizations from the final dataset

Project Repository For the complete automation workflow, implementation details, source code, and documentation, please refer to the GitHub repository: GitHub Repository: https://github.com/NivethaAshokkumarG/Employment-Growth-Analysis-2024-2034-AI-Automation

13. Dashboard Components



13.1 KPI Cards

The Power BI dashboard uses KPI cards to provide a quick summary of the workforce outlook.

The key metrics presented are:

Total Employment — 2024: **Represents the current employment level across the occupations analyzed.**

Total Employment — 2034: **Represents the projected employment level for 2034.**

Employment Change — 2024–2034: **Shows the overall change in employment between the two years.**

Growth Percentage: **Provides an overall view of projected employment growth.**

These KPIs allow users to understand the overall workforce situation before exploring individual occupations.

13.2 Dashboard Visualizations

The dashboard uses visualizations to examine employment concentration and future growth opportunities at the occupation level.

1. Top 10 Occupations by Projected Employment (2034)
A clustered bar chart showing the occupations with the highest projected employment in 2034. This identifies where the largest workforce demand is expected to remain.

2. Occupation Growth Analysis
A visualization showing the distribution/pattern of employment growth based on the available employment-growth measures. This provides an overall view of how occupations are expected to change.

3. Top 10 Occupations by Employment Growth (%)
A bar chart highlighting the occupations with the highest projected percentage growth between 2024 and 2034. This identifies occupations experiencing the fastest relative expansion.

4. Top 10 Occupations by Employment Increase (2024–2034)
A bar chart ranking occupations by absolute employment increase. Unlike percentage growth, this focuses on the actual number of additional jobs projected to be added.

Dashboard Design Approach

The dashboard was designed as a single-page executive view **so that users can quickly move from:**

Overall workforce metrics → Employment concentration → Growth opportunities → Job additions

This provides both a high-level summary and occupation-level insights without requiring multiple dashboard pages.

14. Conclusion

The Workforce Employment & Growth Analysis provides a data-driven view of how employment is expected to change between 2024 and 2034**.The analysis combines employment levels, projected employment, absolute employment change, and percentage growth to identify occupations with significant future workforce opportunities. The Power BI dashboard converts these analytical findings into an interactive business-facing view, allowing users to quickly identify occupations with high projected employment, strong growth rates, and substantial job additions.**

From a business perspective, the analysis can support workforce planning, talent strategy, career planning, and identification of future employment opportunities**. Organizations can use these insights to understand where workforce demand is likely to increase and where employment patterns may require closer attention.**

Overall, the project demonstrates how data preparation, feature engineering, exploratory analysis, and Power BI visualization **can be combined to transform workforce data into actionable business insights.**